μPC610

Polarized 10 Bit D/A Converter

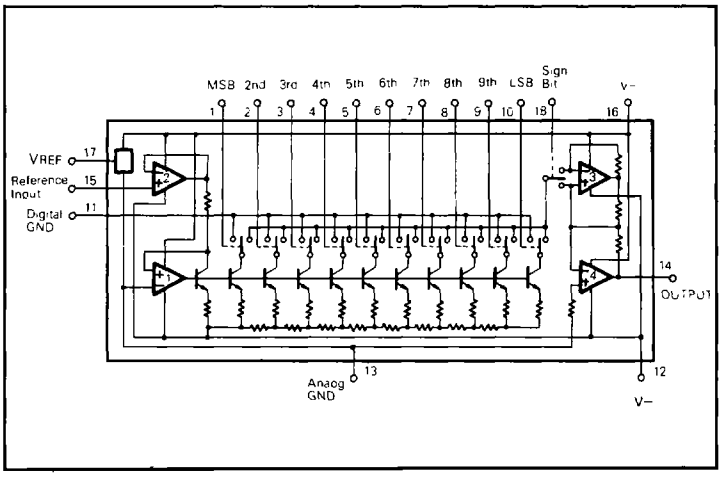
GENERAL DESCRIPTION

The μPC610 is a high performance precision monolithic D/A converter for converting 10 bit binary coded digital signal into an analog DC output voltage. All circuit blocks required for a D/A conversion system are integrated on a single chip, making the composing of a D/A system easy. Futhermore, since the IC is provided with reference voltage and reference input terminals, multiplier type operation is also possible.

FEATURES

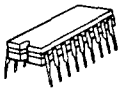
- Full Scale Temp. 100 ppm/°C MAX.
- Linearity Error 0.2% (½ of 8th Bit) MAX.
- Setting Time 1.5 μs TYP.
- Built-in band-gap reference voltage source.
- Multiplying type.
- Sign-Magnitude binary code.
- Low noise.
- Low power dissipation.

EQUIVALENT CIRCUIT



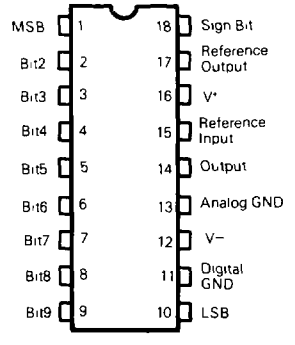
ORDERING INFORMATION

μPC610D



18 pin Ceramic DIP
(Dual In-Line Package)

CONNECTION DIAGRAM (Top View)



μPC610D

ABSOLUTE MAXIMUM RATINGS (Ta = 25°C)

PARAMETER		μPC610	UNIT
Voltage between V^+ and V^-		±18	V
Power Dissipation*	D Package	500	mW
Analog Ground to Digital Ground		±0.5	V
Logic Input Voltage		−5 to +15	V
Reference Input Voltage		0 to +7	V
Reference Voltage Source Output Current		1.0	mA
Output Short Circuit Duration		Indefinite	s
Operating Temperature Range	D Package	−20 to +80	°C
Storage Temperature Range	D Package	−55 to +150	°C

* See thermal information in chapter 11.

ELECTRICAL CHARACTERISTICS ($V^{\pm}$ = ±15 V, Ta = 25°C)

CHARACTERISTIC	MIN.	TYP.	MAX.	UNIT	CONDITIONS
Resolution (Note 1)			11	Bit	Bipolar Operation
			10	Bit	Unipolar Operation
Linearity Error (Note 1)		0.1	0.2	%FSR	Ta = −20 ~ +80°C
Setting Time		1.5	6.0	μs	Final Value ±20 mV
Full Scale Temperature Coefficient (Note 2)		50	100	ppm/°C	Using internal reference voltage source
		30	60	ppm/°C	Using external reference voltage source
Reference Input Bias Current		100	500	nA	
Reference Input Slew Rate		1.5		V/μs	
Reference Voltage	2.2	2.4	2.6	V	$R_L \geqslant 20\ k\Omega$
Zero Scale Offset Voltage		±5	±10	mV	Signbit "ON" Other bits "OFF"
Zero Scale Offset Symmetry		±1	±5	mV	
F.S. Output Offset		±10	±80	mV	
Supply V. Suppression Ratio		0.015	0.15	%FSR/V	$\pm 12\ V \leqslant V^{\pm} \leqslant \pm 18\ V$
Power Dissipation			300	mW	
Logic Input Terminal Current			10	μA	V_{IN} = −5 V ~ +15 V
High Level Input V. (Note 3)	2.0			V	
Low Level Input V. (Note 3)			0.8	V	
Full Scale Output Voltage (Note 4)	10.0		11.0	V	All bits "ON" $R_L \geqslant 2\ k\Omega$
	−11.0		−10.0	V	Signbit "OFF" Other bits "ON" $R_L \geqslant 2\ k\Omega$

Note 1. Though the IC possesses a resolution of 10 or 11 bits, the linearity error is equivalent to 9 bits. In applications where perfect monotonicity is expected, employ the IC as an 8 bit D/A converter.

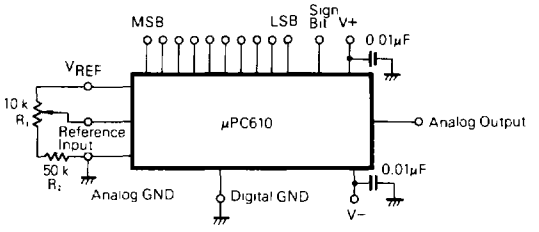
Note 2. The average value of the differential coefficient at Ta = −20°C to +80°C.

Note 3. The digital input is active "High" binary code.

Note 4. The value when the internal reference voltage is directly applied to the reference input terminals.

TYPICAL APPLICATIONS

1. Using internal reference voltage source

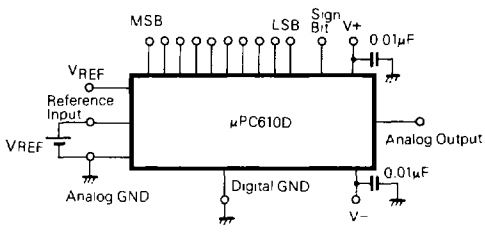


Note: Use resistors with good stability for R_1 and R_2

○ Sign + 10 Bit Binary code

Sign	Input Magnitude	Output
1	1111111111	= 10V
1	0000000000	= 0V
0	0000000000	= 0V
0	1111111111	= -10V

2. External reference and multiplier type



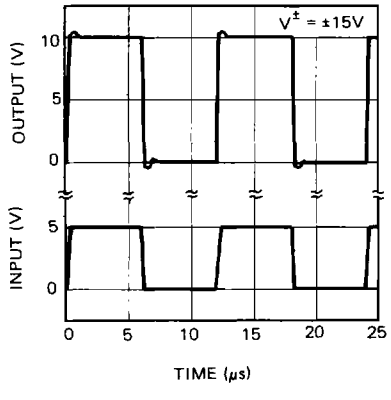
○ Multiplying coefficient ≈ 4.4

$V_{FS} \approx 4.4 \times V_{REF}$, $V_{MSB} \approx 2.2 \times V_{REF}$

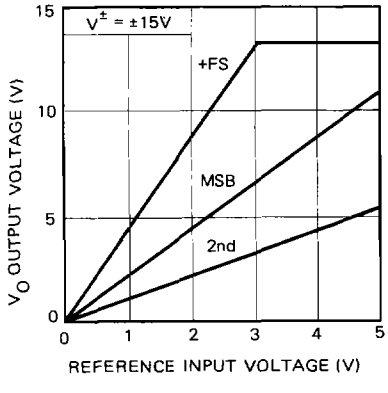
$V_{LSB} \approx 2.2 \times 2^{-9} \times V_{REF}$

TYPICAL PERFORMANCE CHARACTERISTICS (Ta = 25°C)

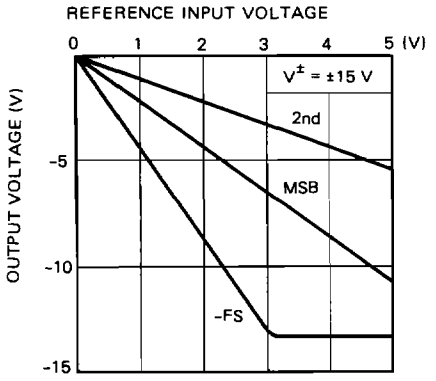
DIGITAL INPUT RESPONSE CHARACTERISTICS



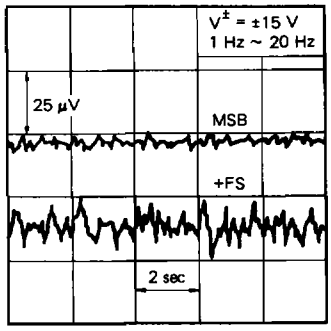
$V_{REF} - V_O{}^+$ CHARACTERISTIC



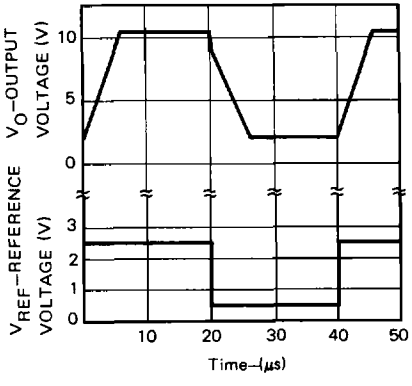
V_{REF} – V_O-CHARACTERISTIC



OUTPUT NOISE VOLTAGE



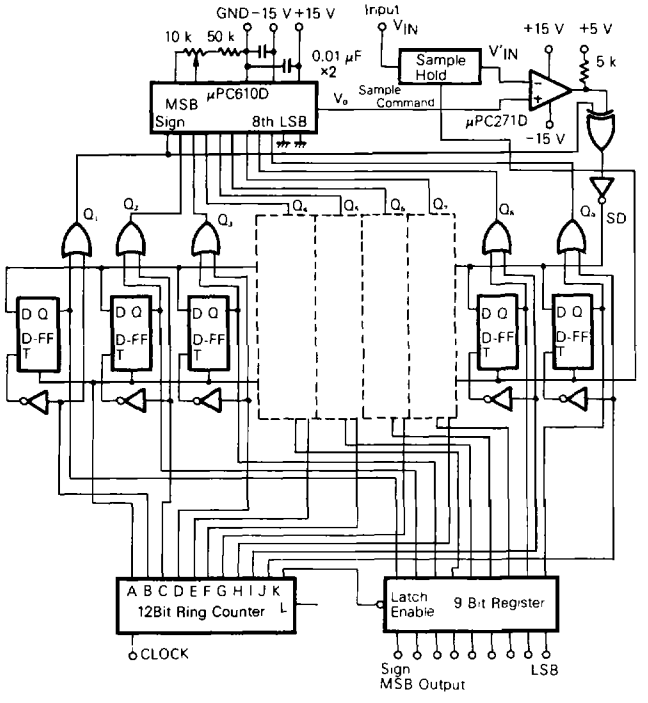
REFERENCE INPUT RESPONSE CHARACTERISTIC

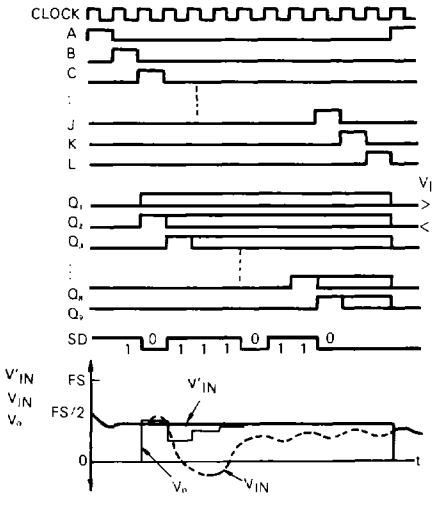


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APPLICATION CIRCUIT

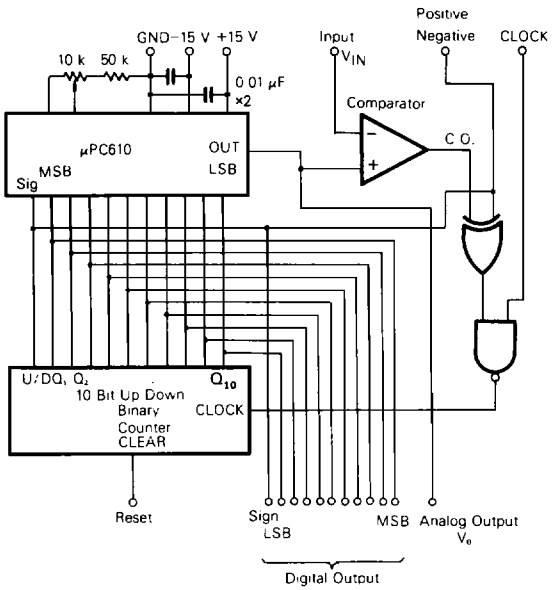
1. 8 Bit + Sign A/D Converter
   Successive Approximation Type

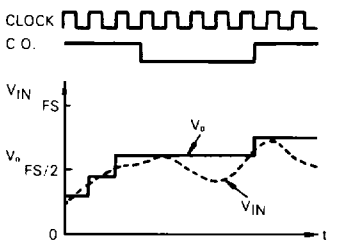




* D, V_{IN}, V_{IN} and V_o show in case when an input equivalent to an output of "101110110"

2. Peak Detector (Positve/Negative)





Sign High

- In case of Sign "H" (Positive)
- The steps at analog outputs are exaggerated from the actual ones.

PRECAUTIONS FOR USAGE

(1) In order to absorb surges and prevent oscillation, bypass the power supply terminals with a capacitor of about 0.01 μF.

(2) In order to utilize the characteristics of the μPC610D in full, employ components of good stability for the full-scale adjustment resistor and the trimmer.

(3) In the case of capacitive loads, since the settling time may increase or oscillation may occur, the μPC610D should be used with load capacitance of 100 pF or less.

(4) In the case of multiplier type operation, the output amplifier will saturate at: $|V_{REF}| \geqslant 3$ V.
In this case the response time and power supply current will increase.

(5) Since the reference potential inside the μPC610D is connected to the analog GND, common mode noise in regards to analog GND will present a direct error.
Since analog GND and digital GND have independent circuits within the IC, when using, (if required), these should be connected together outside the IC.